

Examine how people use data to develop scientific explanations

Learning intention

- examine how people use data to develop scientific explanations.

Teach and model

- explore how hydrologists use rainfall and water use data to explain the amount of water flowing in rivers and why this changes over time.
- viewing or listening to documentaries or news reports that feature researchers and identifying how they talk about their area of research.
- investigating how ecologists use food chain data to develop explanations for population decline of native species such as the Richmond birdwing butterfly.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.